Ch - 02 STRUCTURE OF ATOM

Que :- What are the fundamental particle of the atom ?

Ans :- There are three fundamental particle of atom.

1. Electron (e)
2. Protone (p)
3. Neutron (n)

Que :- Write the properties of electron, protone & neutron.

Ans :- Electron :- They are negatively charged. They are present in the outer shells within an atom and orbit the positively charged nucleus in well-defined orbits. The mass of an electron is about 1/2040 times the mass of a hydrogen atom. An electron is represented as "e".

Proton :- They are positively charged. They are present in the nucleus of all atoms. The mass of the proton is take as one unit and equals the mass of a neutron

A proton is represented as "p".

Neutron :- They are neutral. They are present in the nucleus of all atoms expect hydrogen. The mass of the neutron is considered as one unit and it equals the mass of a proton. A neutron is represented as "n".

Property	Electron	Proton	Neutron
Location	Extra nuclear portion (orbit)	Inside nucleus	Inside nucleus
Absolute Charge	-1.6×10^{-19} C	$+1.6 \times 10^{-19}$ C	Nil
Relative charge	-1	$+1$	Nil
Absolute mass	9.01×10^{-31} kg	1.67×10^{-27} kg	1.675×10^{-27} k
Relative mass	1/1840	1	1
Discover	J.J. Thomson (1897)	E. Goldstein (1886)	James Chedwick (1932)

Page No. :
Date : / /

- **Atomic Number** :- The number of the unit positive charge present in the nucleus of atom is known as atomic number of the elements. It is denoted by "Z".

Atomic no. (Z) = No. of protons (p) = No. of electron (e)

$$\boxed{Z = p = e}$$

- **Mass Number** :- The sum of the number of protons and neutrons present in the nucleus of an atom is called the mass number. It is denoted a "A".

Mass no. (A) = No. of protons (p) + No. of neutrons (n)

$$\boxed{A = p + n}$$

- **Isotopes :-** The isotopes are the atoms of the same element which have the same atomic number but different mass number.

For ex - H^1 = Protium
$_1H^2$ or $_1D^2$ = Deuterium
$_1H^3$ or $_1T^3$ = Tritium

~~Isopo~~ Isotopes of Carbon -

C_6^{12}, C_6^{14}

- **Isobars :-** Isobars are the atoms of different elements which have same mass number but different atomic number.

For ex :- $_{18}Ar^{40}$, K_{19}^{40}, Ca_{20}^{40}

- **Isotones :-** Isotones are the atoms of different elements which content same number of neutrons.

For ex - C_6^{14}, N_7^{15}, O_8^{16}

All these have 8 neutrons.

Page No. :
Date : / /

Ques1. – Calculate the no. of protons, neutrons, electrons in $_{35}Br^{80}$.

Ques2. – The atomic no. of an element is 5 and mass no. is 11. find no. of protons, neutrons and electrons present in an atom of it. How can these element represented ?

Ques3. – There are 14 protons & 13 neutrons in a nucleus of an atom. What is its mass number ?

Ques4. – Indicate the no. of electrons, protons & neutrons ~~present in~~ in the element K^{39}_{19}.

solve(1) $Z = 35$

$Z = p = e = 35$

$A = p + n$

$n = A - p$

$n = 80 - 35$

$\boxed{n = 45}$

solve(2) $Z = 5$

$Z = p = e = 5$.

$n = A - p$

$n = 11 - 5$

$\boxed{n = 6}$

Page No.:
Date: / /

solve (3)

$A = p + n$

$A = 14 + 13$

$\boxed{A = 27}$

solve (4)

$Z = 19$

$Z = p = e = 19$

$n = A - p$

$n = 39 - 19$

$\boxed{n = 20}$

Que :- Describe Rutherford model.

Ans :- In 1911, Rutherford performed α-ray scattering experiment to know about the position of the two fundamental particles electrons and protons in an atom.

From the experiment, Rutherford made the following observations

(i) Most of the α-particles (about 99%) passed through the gold foil undeflected

(ii) Some of these particles were deflected by small angles

(iii) A very few α-particles (one out of about 20000 particles) suffered major

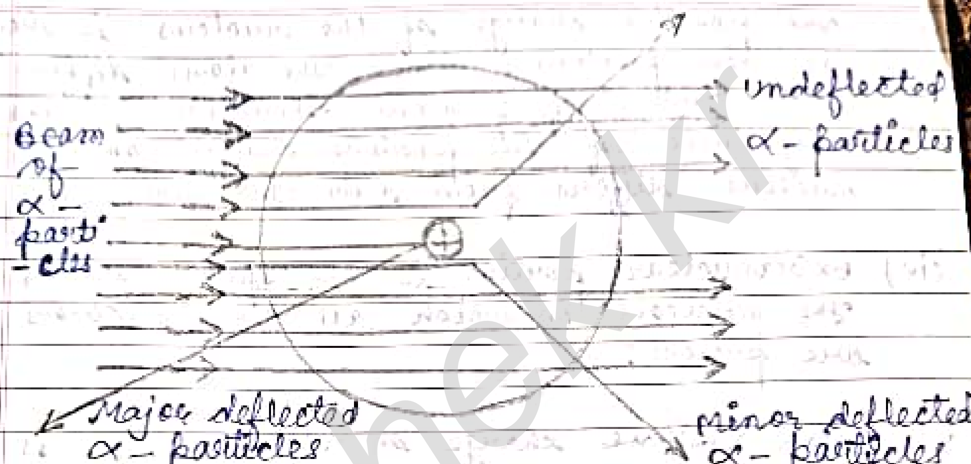
Page No. :
Date : / /

deflection by more than 90° and even come back in the same direction.

Conclusions from the Observations

(i) As most of the α-particles passed through the atoms of gold undeflected, this means that they did not come across any obstruction in their path. Thus, most of the space inside an atom is expected to be empty. The electrons with negligible mass were supposed to be present in the space.

(ii) As a few α-particles suffered minor deflections and a very few even major deflections, this means that they must have come across some obstruction in their path which must be very small (only a few particles were deflected), heavy (each α-particle has 4 units mass) and positively charged (each α-particle carries +2 units charge).



Rutherford's Model of Atom

In the light of the α-ray scattering experiment, Rutherford gave the following picture of the atom.

(i) An atom consists of two parts known as nucleus and extra nuclear portion.

(ii) Nucleus is present in the centre of the atom. It is massive positively charged and also extremely small in size. The radius of the nucleus is about 10^{-15} m while that of the atom is about 10^{-19} m. Thus, the size of the nucleus is very small as compared to that of the atom

Page No. :
Date : / /

(iii) The positive charge of the nucleus is due to the protons. Since the atoms differ in the number of protons therefore, the magnitude of the positive charge on the nucleus differs from atom to atom.

(iv) Extranuclear portion is the space around the nucleus in which all the electrons are present.

(v) Total positive charge on the nucleus is equal to the total negative charge on the electrons and the atom as a whole is electrically neutral.

(vi) Electrons in the extra nuclear portion are not stationary but are revolving around the nucleus at high speeds in circular paths called orbit. Revolving electrons have certain centrifugal force acting away from the ~~nu~~ nucleus which balances the force of attraction directed towards the nucleus.

Page No. :
Date : / /

(iii) The positive charge of the nucleus is due to the protons. Since the atoms differ in the number of protons therefore, the magnitude of the positive charge on the nucleus differs from atom to atom.

(iv) Extranuclear portion is the space around the nucleus in which all the electrons are present.

(v) Total positive charge on the nucleus is equal to the total negative charge on the electrons and the atom as a whole is electrically neutral.

(vi) Electrons in the extra nuclear portion are not stationary but are revolving around the nucleus at high speeds in circular paths called orbit. Revolving electrons have certain centrifugal force acting away from the ~~nu~~ nucleus which balances the force of attraction directed towards the nucleus.

Name of Elements	Symbol	Atomic Number
Hydrogen	H	1
Helium	He	2
Lithium	Li	3
Beryllium	Be	4
Boron	B	5
Carbon	C	6
Nitrogen	N	7
Oxygen	O	8
Fluorine	F	9
Neon	Ne	10
Sodium	Na	11
Magnesium	Mg	12
Aluminium	Al	13
Silicon	Si	14
Phosphorus	P	15
Sulfur	S	16
Chlorine	Cl	17
Argon	Ar	18
Potassium	K	19
Calcium	Ca	20
Scandium	Sc	21
Titanium	Ti	22
Vanadium	V	23
Chromium	Cr	24

Page No.
Date : / /

Electronic Configuration

(1) digit method

(2) s, p, d, f

$\boxed{2}$ $\boxed{8}$ $\boxed{18}$ $\boxed{18}$ $\boxed{18}$

I II III IV V

Na(11) → 2, 8, 1 valence electron = 1

Core electron V.E

10

Mg (12) → 2, 8, 2

He(2) → 2

s → 2

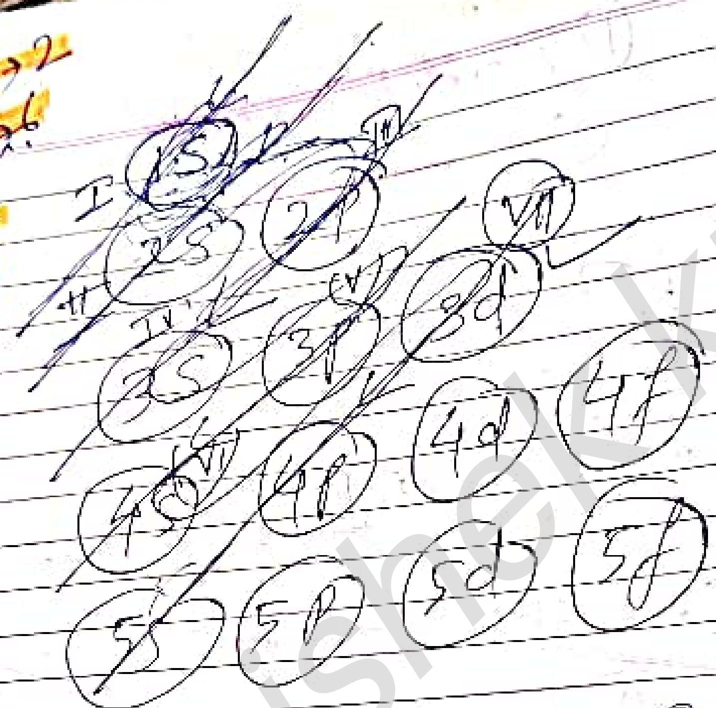
p → 6

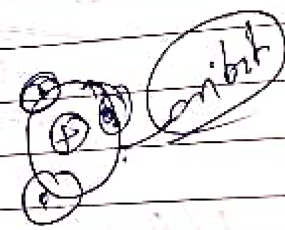
d → 10

f → 14

Page No.:
Date: / /

s → 2
p → 6
d → 10
f → 14





H(1) → $1s^1$

He(2) → $1s^2$

Li(3) → $1s^2 2s^1$

Na(11) → $1s^2 2s^2 2p^6 3s^1$

Ca(20) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

Sc(21) →

Electron Configuration

Digit method

1. H (1) → 1
   Valence electron → 1 , Core electron → 0

2. He (2) → 2
   V.E → 2 , Core C.E → 0

3. Li (3) → 2 , 1
   V.E → 1 , C.E = 2

4. Be (4) → 2 , 2
   V.E → 2 , C.E → 2

5. B (5) → 2 , 3
   V.E → 3 , C.E → 2

6. C (6) → 2 , 4
   V.E → 4 , C.E → 2

7. N (7) → 2 , 5
   V.E → 5 , C.E → 2

8. O (8) → 2 , 6
   V.E → 6 , C.E → 2

Page No. :
Date : / /

27. Co (27) → 2, 8, 8, 9
V.E → 9 , C.E → 18

28. Ni (28) → 2, 8, 8, 10
V.E → 10 , C.E → 18

29. Cu (29) → 2, 8, 8, 11
V.E → 11 , C.E → 18

30. Zn (30) → 2, 8, 8, 12
V.E → 12 , C.E → 18

s, p, d, f form

1. H (1) → $1s^1$
2. He (2) → $1s^2$
3. Li (3) → $1s^2 2s^1$
4. Be (4) → $1s^2 2s^2$
5. B (5) → $1s^2 2s^2 2p^1$
6. C (6) → $1s^2 2s^2 2p^2$
7. N (7) → $1s^2 2s^2 2p^3$
8. O (8) → $1s^2 2s^2 2p^4$
9. F (9) → $1s^2 2s^2 2p^5$
10. Ne (10) → $1s^2 2s^2 2p^6$
11. Na (11) → $1s^2 2s^2 2p^6 3s^1$
12. Mg (12) → $1s^2 2s^2 2p^6 3s^2$

Page No.:
Date: / /

13. Al (13) → $1s^2 2s^2 2p^6 3s^2 3p^1$
14. Si (14) → $1s^2 2s^2 2p^6 3s^2 3p^2$
15. P (15) → $1s^2 2s^2 2p^6 3s^2 3p^3$
16. S (16) → $1s^2 2s^2 2p^6 3s^2 3p^4$
17. Cl (17) → $1s^2 2s^2 2p^6 3s^2 3p^5$
18. Ar (18) → $1s^2 2s^2 2p^6 3s^2 3p^6$
19. K (19) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
20. Ca (20) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
21. Sc (21) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$
22. Ti (22) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
23. V (23) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$
24. Cr (24) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$
25. Mn (25) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
26. Fe (26) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$
27. Co (27) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
28. Ni (28) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$
29. Cu (29) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$
30. Zn (30) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$

Que:- Write electronic configuration of
(1) Na^+, (2) Mg^{2+}, (3) Cl^-, (4) Cr^{3+}, (5) Fe^{++}, (6) Fe^{3+},
(7) Cu^{++}, (8) Cu^{2+},

(+) ⟶ (−)
(−) ⟶ (+)

Page No.: Date: / /

1. Na^{+} (10) → $1s^2\ 2s^2\ 2p^6$
2. Mg^{2+} (10) → $1s^2\ 2s^2\ 2p^6$
3. Cl^{-} (18) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$
4. Cr^{3+} (21) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$
5. Fe^{++} (24) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^4$
6. Fe^{3+} (23) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^3$
7. Cu^{++} (27) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^7$
8. Cu^{3+} (26) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$

Que :- What are the difference between orbit and orbital

Ans :-

Orbit	Orbital
(i) An orbit is the well defind circullar path around the neucleus which the electron revolves.	An orbital represents the region in space around the neucleus of the atom where the probabilty of finding the electrones the maxium (90 - 95%)
(ii) It represents the movement of the electron around the neucleus in a plan.	(ii) It represents the three dimensional movement of the electron.
(iii) All orbits are circular.	(iii) All orbitals are different shapes.

(iv) The maximum no. of electrons which can be accomodorate in the an orbit is $2n^2$ (where n = no. of the orbit).

accommodate

(iv) An orbital can not accomodate more than two electrons.

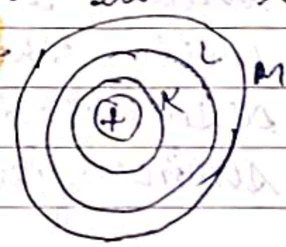
Que :- What are quantum numbers ? Explain with example.

Ans :- ~~Quatum~~ Quantum no. :- The set of four no. [Principle Quantum no. (n), Azimuthal Quantum no. or Angular Quantum no. (l), Megenatic Quantum no. (m_l), Spin Quantum no. (s)] which give a complete information about the electron in an atom i.e its energy, shape, orientation and also its spin.

$$\boxed{E_n = \frac{-13.6}{n^2}}$$

energy levels unit

eV (electro Volt)



- **Principle Quantum Number (n)** :- The quantum no. was given by Bohr and its denoted by n. It determines the main energy level or shell in which an electron in an atom ~~is~~ is present and also the energy associated with it. In addition it also

Page No.:
Date: / /

determines the average distance of the electron from the nucleus in particular shape. Starting from the nucleus, the energy shells are denoted K, L, M, N ... or 1, 2, 3, etc. The maximum no. of electron which a shell can ~~accomodorate~~ accommodate is $2n^2$. Thus, K-shell $(n=1) = 2$ electrons. L-shell $(n=2) = 8$ electrons. M $(n=3) = 18$ electron, N $(n=4) = 32$ electrons.

- **Azimuthal or subsidiary and angular Quantum number (l)** :- Based on the energy of electron the shells are divided into sub-levels or sub-shells and these have been describe with the help of this quantum no. designated has (l). It's given by Sommerfield.

The value of l depends upon the value of shell (n) with which it is associated. For a particular value of n, l can have all integer values ranging from 0 to (n-1). Each value of l represent one particular ~~substance~~ sub-shell.

$n = 1$ (K)

$l = n - 1 = 1 - 1 = 0$; l has 0 (one sub-shell)

$n = 2$ (L)

L-shell $= 2 - 1 = 1$; l has 0, 1 (two sub-shells)

$n = 3$ (M)

$l = 3 - 1 = 2$; l has 0, 1, 2 (three sub-shells)

Value of n	Value of l	Name of the sub-shell	No. of sub-shells
1.	0	1s	one
2.	0, 1	2s, 2p	two
3.	0, 1, 2	3s, 3p, 3d	three
4.	0, 1, 2, 3	4s, 4p, 4d, 4f	four

Azimuthal quantum number also gives the orbital angular momentum (L) of the electron present in any sub-shell and also the shapes of the orbitals present in a particular sub-shell. Angular momentum of the electron is given as :

$$\sqrt{l(l+1)}\ \frac{h}{2\pi}$$

For s-orbital $l = 0$; $L = \sqrt{0(0+1)} = (h/2\pi) = 0$.

For p-orbital $l = 1$; $L = \sqrt{1(1+1)} = (h/2\pi) = \sqrt{2}\ h/2\pi$

For d-orbital $l = 2$; $L = \sqrt{2(2+1)} = (h/2\pi) = \sqrt{6}\ h/2\pi$

For f-orbital $l = 3$; $L = \sqrt{3(3+1)} = (h/2\pi) = \sqrt{12}\ h/2\pi$

Page No. :
Date : / /

- **Magnetic Quantum No. (m_l) :-** When a source producing a spectral line is placed in a magnetic field, a spectral line splits into a no. of lines. This is called Zeeman's effect. In order to explaine this effect Linde a proposed that an electron because of its angular motion around the nucleus produces an electric field which in turn leads to a magnetic field. The magnetic field of electron is likely ~~ligh~~ ~~lightly~~ to interact with the magnetic field of the earth. As a result, the electron present in a particular sub-shell acquire certain specific orientations in space called orbital. These orientation are specified by a quantum no. called magnetic quantum no. denoted as 'm'. Thus, each value of m corresponds to a specific orbital. The value of 'm' are linked to that of l (azimuthal quantum no.) therefore magnetic quantum no. is quite often denoted as {m_l}. for a given value of l, the possible value of m_l vary from $-l$ to 0 and ~~to~~ to $+l$, the total value of m_l are $(2l+1)$.

Value of l	Name of sub-shell	Value of m or (m_l) $[m = 2l+1]$	Name of the orbital
$l=0$	s-sub shell	$m = 0$ to 0	s-orbital (one)
$l=1$	p-sub shell	$m = -1, 0, +1$	p-orbital (three)
$l=2$	d-sub shell	$m = -2, -1, 0, +1, +2$	d-orbital (five)
$l=3$	f-sub shell	$m = -3, -2, -1, 0, +1, +2, +3$	f-orbital (seven)

s - sharp
p - principle
d - defuse
f = fundamental

- **Spin Quantum No. (s)**: - An electron besides charge & mass also spins around its axis & this leads to quantum no. called spin quantum no. (s). Which was given for the first time by Goudsmit. This quantum no. as ~~said~~ stated earlier, does not follow from the wave machinical model ~~method~~ which gives rise to the other three quantum numbers, since the electron can spin either in the clockwise or anticlockwise direction, s can have two values. $+\frac{1}{2}$ & $-\frac{1}{2}$ ↑ (spin up) & ↓ (spin down).

Page No. :
Date : / /

Que (i) :- An atomic orbital has $n = 3$ what are the possible value of l ?
(ii) An atomic orbital has $l = 3$ what are the possible value of m_l ?

(i) $n = 3$
$l = 3 - 1 = 2$
$0, 1, 2$

(ii) $l = 3$

$m_l = 2l + 1$
$= 2(3) + 1$
$= 6 + 1 = 7$

$-3, -2, -1, 0, +1, +2, +3$

Que :- Using s, p, d notations describe the orbitals with following quantum no.

a) $n = 1, l = 0$ — 1s orbital
b) $n = 2, l = 0$ — 2s orbital
c) $n = 3, l = 1$ — 3p orbital
d) $n = 4, l = 2$ — 4d orbital

Que :- An electron in present 4f sub-shell write the possible values for 4f quantum no.

solve principle quantum no.

$(n) = 4$

$l = 3$

and $m_l = 7 = -3, -2, -1, 0, +1, +2, +3$

$s = +\frac{1}{2}, -\frac{1}{2}$

— write short notes on

- **Aufbau Principle**

— The word Aufbau in german means building up and this related to ~~building~~ filling up the electrons in the different orbital.

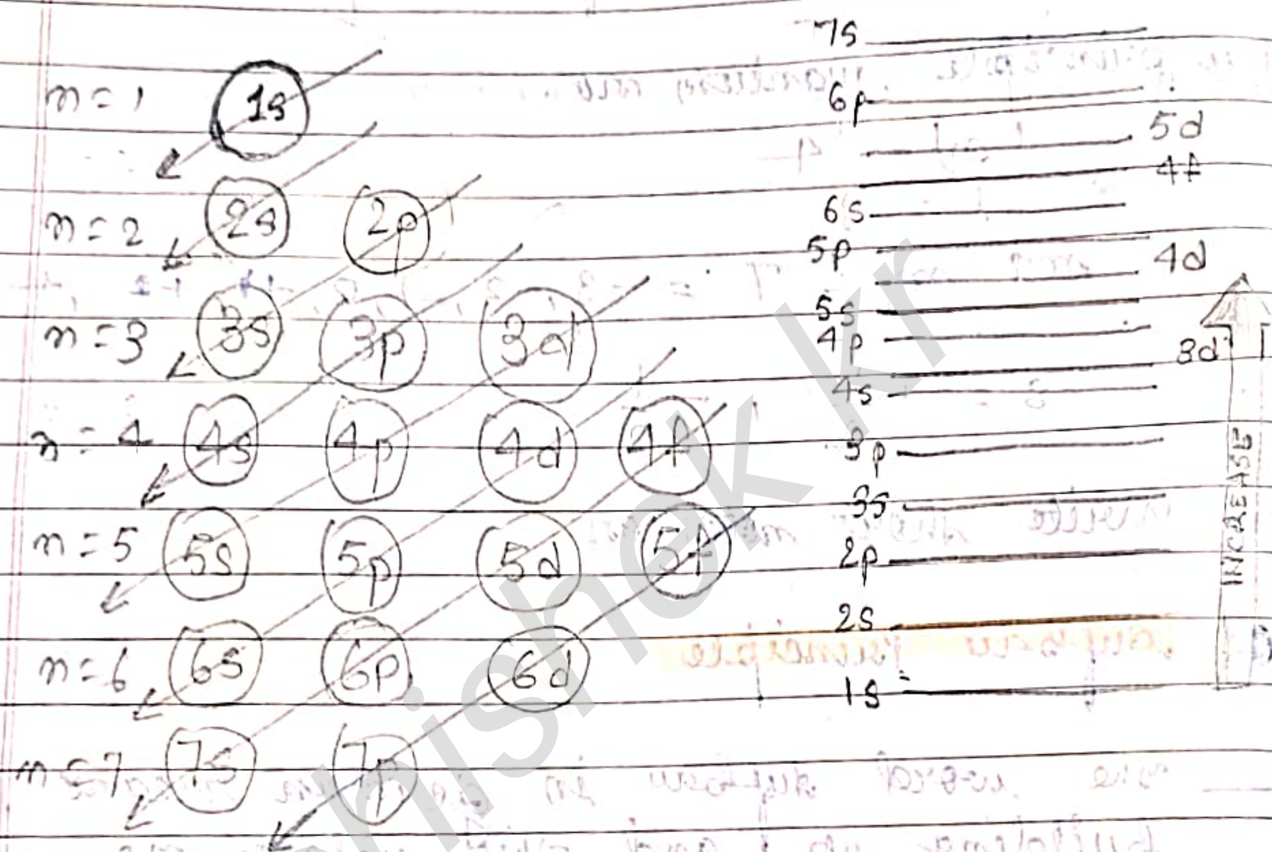
According to aufbau principle

~~at~~ In the ground states of an atom, the electrons are added progressively to the various orbitals in increasing order of energies starting from the orbital of lowest energy.

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p,

Page No. :
Date : / /

5s, 4d, 5p, 6s, 4f, 5d.



- **Hund's rule of maximum multiplicity**

When more than one orbitals of equal energies are available, the electrons will first occupy these orbitals singly with parallel spins. The pairing of electrons will then take place.

Page No. :
Date : / /

Atomic No (Z)	Element	Electronic configuration	1s	2s	2p	3s
1.	Hydrogen (H)	$1s^1$	↑			
2.	Helium (He)	$1s^2$	↑↓			
3.	Lithium (Li)	$1s^2 2s^1$	↑↓	↑		
4.	Beryllium (Be)	$1s^2 2s^2$	↑↓	↑↓		
5.	Boron (B)	$1s^2 2s^2 2p^1$	↑↓	↑↓	[↑][ ][ ]	
6.	Carbon (C)	$1s^2 2s^2 2p^2$	↑↓	↑↓	[↑][↑][ ]	
7.	Nitrogen (N)	$1s^2 2s^2 2p^3$	↑↓	↑↓	[↑][↑][↑]	
8.	Oxygen (O)	$1s^2 2s^2 2p^4$	↑↓	↑↓	[↑↓][↑][↑]	
9.	Fluorine (F)	$1s^2 2s^2 2p^5$	↑↓	↑↓	[↑↓][↑↓][↑]	
10.	Neon (Ne)	$1s^2 2s^2 2p^6$	↑↓	↑↓	[↑↓][↑↓][↑↓]	
11.	Sodium (Na)	$1s^2 2s^2 2p^6 3s^1$	↑↓	↑↓	[↑↓][↑↓][↑↓]	↑
12.	Magnesium (Mg)	$1s^2 2s^2 2p^6 3s^2$	↑↓	↑↓	[↑↓][↑↓][↑↓]	↑↓

Page No. :
Date : / /

Pauli's Exclusion Principle

No two electrons in an atom can have all the four quantum numbers

Pauli in 1925, has given a very important principle known as Pauli's exclusion principle. according to it, No two electrons in an atom can have all the four quantum numbers same. Two electrons present in an atom can have ~~all the~~ a maximum of three quantum numbers same ie they can be present in the same shell, the same sub-shell and same orbital. However, they must have opposite spins i.e the spin quantum numbers are different. If one spins in the clockwise direction, the other must have anticlockwise spin.

In order to illustrate the principle, let us consider two electrons present in the K-shell. For them;

$n=1$	$l=0$	$m=0$	$s=+1/2$ & $-1/2$
(same)	(same)	(same)	(different)

Page No. :
Date : / /

Thus, it is quite clear that an orbital can have a maximum of two electrons ~~present~~ with opposite spins. The orbital with two electrons is called filled orbital while the orbital having only one electron is known as half filled orbital.

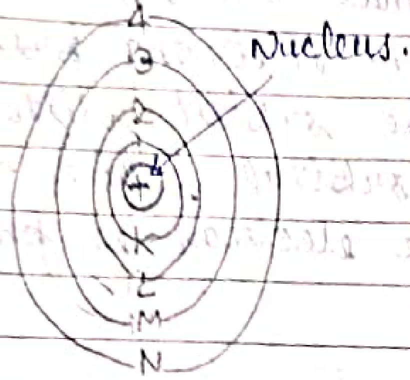
Que :- Write down Bohr' model of an atom.

(i) In a hydrogen atom, the electron is revolvin around the nucleus in a well defined circular path of fixed radius and energy. The circular path is known as ~~orbit~~ and the ~~corr~~ corresponding energy state of the electron is called stationary state.

(ii) The energy of an electron in its orbit does not change of its own with time.

(iii) When a required ~~amou~~ amounts of ~~bundles~~ called energy is absorbed by the electron it absorbs the same in fixed amounts of bundles called quanta and jumps to higher stationary state called excited state. The same energy is released when the electron jumps from higher stationary state to the lower stationary state. Mathematical

$$\Delta E = E_2 - E_1$$



Please remember that the electronic energy in a particular energy state, also called energy level is stationary. But the electron is not stationary.

(iv) The different energy levels or energy shells for the electrons are designed as K, L, M, N..... etc. or as 1, 2, 3, 4 ..... etc.

(v) Only those energy orbits are permitted for the electron in which the angular momentum of the electron is a whole number multiple of $\frac{h}{2\pi}$ i.e.

Angular momentum of electron $(mvr) = \frac{nh}{2\pi}$.
(n = 1, 2, 3, 4 ..... etc.) Here m is the mass of the electron, v is the tangential velocity of the revolving electron, r is the radius of the orbit in which electron revolves, h is Planck's constant and n is an integer. Thus, the angular momentum of the electron can be $h/2\pi$ $(n=1)$, $2h/2\pi$ $(n=2)$, $3h/2\pi$ $(n=3)$ ..... etc. This means that the angular momentum of the electron is quantised.

Page No.:
Date: / /

(vi) The most important property of an electron is the energy of its stationary state. It is given by the expression:

$$E_n = -R_H\left(\frac{1}{n^2}\right);\ n = 1, 2, 3, \ldots \text{ etc}.$$

Here R_H is called Rydberg's constant and its value is 2.18×10^{18} J.

Heisenberg's Uncertainty Principle

— In 1927, Heisenberg's uncertainty principle. It may be stated as follows:

It is not possible to measure simultaneously the ~~exact~~ position and the ~~exact~~ momentum (velocity) of ~~the~~ a microscopic particle with absolute accuracy or certainty.

Alternatively, the principle may also be stated as:

The product of uncertainty in position and uncertainty in momentum of a microscopic particle is always constant and is equal to or greater than $h/4\pi$.

Page No. :
Date : / /

Mathematically, $\Delta x . \Delta p \geq h/4\pi$
Here, Δx = uncertainty in measuring exact position
Δp or $m\Delta v$ = uncertainty in measuring exact momentum or velocity

For solving problems, the mathematically expression of uncertainty principle is generally written as:

$$\Delta x . \Delta p = \frac{h}{4\pi} \quad \text{or} \quad \Delta x . m\Delta v = \frac{h}{4\pi}$$

Que:- Calculate the uncertainty in the momentum of the an electron if it is confinded to a linear reason of length 1×10^{-10} m.

Sol: According to uncertainty principle,

$$\Delta x . \Delta p = \frac{h}{4\pi}$$

$$\Delta p = \frac{h}{4\pi . \Delta x}$$

$$\Delta p = \frac{6.626 \times 10^{-34}}{4 \times 3.14 . 10^{-10}}$$

$$\Delta p = \frac{6.626 \times 10^{-34+10} \times 100}{12.56 \times 100} \times \frac{10}{\times 10}$$

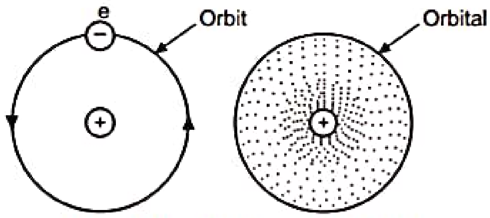


Fig. 1.11 : Orbit and Orbital

The fixed orbits of Bohr's atom are now replaced by orbitals and each orbital can contain a maximum of two electrons with opposite spins.

From the above discussion about sub-energy levels (sub-shells) s, p, d, f contain 2, 6, 10, 14 as the maximum number of electrons. Hence, there is 1s-orbital, 3p orbitals, 5d orbitals and 7f orbitals. If the principal quantum number is 'n', there are $2n^2$ electrons and 'n^2' orbitals in the main energy level.

s - Orbitals

For each main energy level, there is an s-orbital. For n = 1, there is only ($n^2 = 1^2$) = 1s orbital. Electrons moving in 's' orbital are known as 's' electrons and are represented as $1s^2$ or $2s^2$ or $3s^2$ etc. The s-orbitals are spherically symmetrical around the nucleus and hence they are non-directional in character. For each value of 'n' there is one s-orbital and there will be 'ns' orbitals, which are concentric spherical regions increasing progressively away from the nucleus. In between the two spheres, the space is empty and capacity of each s-orbital is of two electrons.

p - Orbitals

For n = 2, there are in all four orbitals ($n^2 = 2^2 = 4$) in second (L) energy level. Out of them, one is 2s and three are p-orbitals. All the p-orbitals are of equal energy except that in magnetic field. Each p-orbital is dumb-bell shaped. It consists of two lobes with an atomic nucleus in between the two lobes, where there is practically zero probability of finding the electron (nodal plane). The three p–orbitals are designated as p_x, p_y and p_z. Each 'p' orbital contains maximum of two electrons present in two lobes. They are oriented along the three mutually perpendicular axes x, y, z respectively. Such orbitals which differ in orientation but having equal energy are called '*degenerate orbitals*'. Therefore, p orbitals have directional character.

The shapes of 's' and 'p' orbitals are shown in Fig. 1.12 (a) and (b).

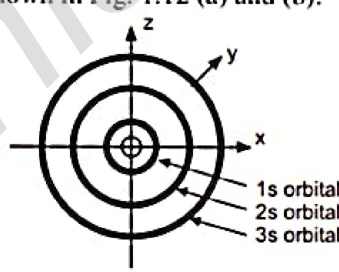


Fig. 1.12 (a) : s - orbital

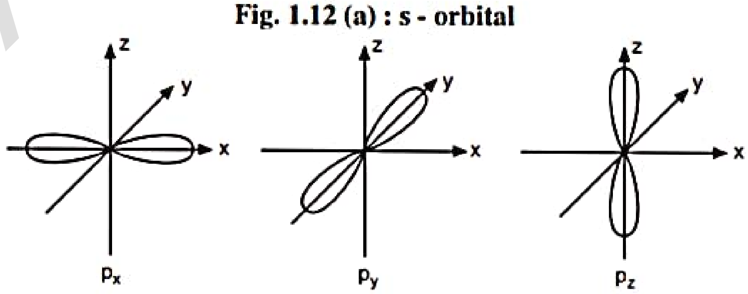


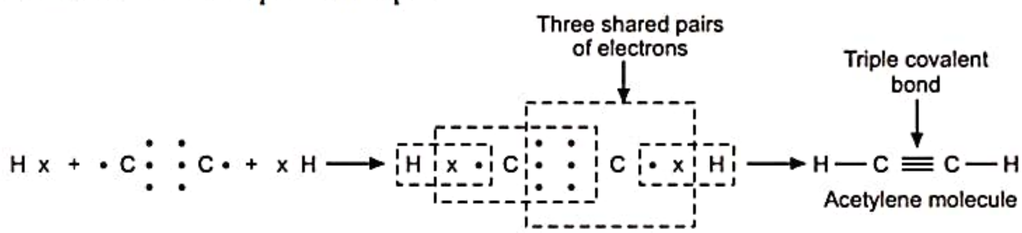
Fig. 1.12 (b) : p-orbitals (Relative orientations of p_x, p_y, p_z orbitals)

d and f orbitals

For n = 3, the total number of orbitals ($n^2 = 3^3 = 9$). Out of these 1 → 3s; 3 → 3p; remaining 5 → 3d orbitals.

For n = 4, the total number of orbitals is $n^2 = 4^2 = 16$. Out of these 1 → 4s, 3 → 4p; 5 → 4d and remaining 7 → 4f orbitals, the geometrical shapes of d and f orbitals are somewhat complicated.

(b) Formation of acetylene molecule (C_2H_2) : Acetylene molecule consists of two atoms of carbon (2, 4) and two atoms of hydrogen (1). Each atom of carbon is in short of 4 electrons to complete the octet, while each hydrogen atom is in short of 1 electron to complete the duplet.



Thus, acetylene molecule is formed by sharing three pairs of electrons i.e. forming triple covalent bond between two carbon atoms and one pair of electrons i.e. single covalent bond with each hydrogen atom.

Properties of covalent compounds :

1. Covalent compounds are generally insoluble in water (polar solvent), but readily soluble in organic solvents such as benzene, carbon tetrachloride (non-polar solvents).
2. Covalent compounds do not ionise when dissolved in water. Thus, they are non-polar in nature. Hence, they do not conduct electricity.
3. Covalent compounds are generally volatile in nature and usually have low melting and boiling points.
4. These are generally organic compounds.

1.28 DISTINCTION BETWEEN ELECTROVALENT AND COVALENT COMPOUNDS

Sr. No.	Electrovalent Compounds	Sr. No.	Covalent Compounds
1.	These are formed by loss and gain of electrons between dissimilar atoms.	1.	These are formed by mutual sharing of electrons between similar or dissimilar atoms.
2.	They are found to exist in the form of ions even in the solid state.	2.	They are not found to exist in the form of ions in the solid or liquid state.
3.	These are polar compounds. Thus, when melted or dissolved in a solvent, they are ionised and hence conduct electric current.	3.	These are non-polar compounds. Thus, when melted or dissolved in a solvent, they are generally not ionised and hence do not conduct electric current.
4.	They possess comparatively high melting and boiling points.	4.	They possess comparatively low melting and boiling points.
5.	They are generally non-volatile and insoluble in organic solvents.	5.	They are generally volatile and usually soluble in organic solvents.

EXERCISE

1. Write short answers of the following : [2 marks]
   (a) Why is an atom electrically neutral ?
   (b) (i) The nucleus consists of 9 protons and 10 neutrons. What should be its atomic weight and electrochemical nature ?
   (ii) For an element having atomic number 9 and mass number 19, write number of protons and neutrons in it.
   (c) State the number of sub-shells in K, L, M, N shells.
   (d) State the relation between atomic number, mass number and neutrons in an atom.
   (e) Define atomic orbital.
   (f) Why is the nucleus of an atom positively charged ?
   (g) Calculate the atomic number and atomic mass number of an atom containing 19 electrons and 20 neutrons.

1.23 TYPES OF VALENCY

There are three types of valency :

(1) Electrovalency (or Electrovalent bond or Ionic bond)

(2) Co-valency (or Covalent bond or non-ionic bond)

(3) Co-ordinate valency. (Not in syllabus)

1.24 ELECTROVALENCY

"The number of electrons that an atom of an element gains or losses to complete its last orbit is called electrovalency."

In this type of valency, one atom loses certain number of valency electrons and attains a stable configuration (with 8 electrons in the last shell), thus acquiring a positive charge. The other atom gains or accepts the same number of electrons and completes its octet, thus acquiring the same negative charge. This loss and gain of electrons takes place in dissimilar atoms to complete their respective orbits. Since, the two atoms have developed charges due to gaining and losing of electrons, therefore, they are held together by electrostatic forces of attraction.

Thus, a bond or linkage is formed between the two dissimilar atoms or ions due to electrostatic forces of attraction, which is called *'electrovalent bond'* or electrovalent linkage or ionic bond.

Types of Electrovalency : Electrovalency is of two types :

(a) Positive electrovalency : The valency obtained by the loss of valency electrons from the atom of metallic element so as to complete its last shell is known as positive electrovalency. For example

(i) Sodium atom losses *'one valency electron'* to complete its octet. Thus, positive electrovalency of sodium atom is +1 (Na^+).

$$\underset{(2,\,8,\,1)}{Na} \longrightarrow \underset{(2,\,8)}{Na^+} + e^-$$

(ii) Magnesium atom losses *'two valency electrons'* to complete its octet. Thus, positive electrovalency of magnesium atom is +2 (Mg^{++}).

$$\underset{(2,\,8,\,2)}{Mg} \longrightarrow \underset{(2,\,8)}{Mg^{++}} + 2\bar{e}$$

(iii) Aluminium atom losses *'three valency electrons'* to complete its octet. Thus, positive electrovalency of aluminium atom is +3 (Al^{+++}).

$$\underset{(2,\,8,\,3)}{Al} \longrightarrow \underset{(2,\,8)}{Al^{+++}} + 3e^-$$

(b) Negative electrovalency : The valency obtained by the *'gain of electrons'* by the atoms of non-metallic elements, so as to complete their octets is known as negative electrovalency.

Examples :

(i) Chlorine atom gains *'one valency electron'* so as to complete its last shell. Hence, the negative electrovalency of chlorine is –1 (Cl^-).

$$\underset{(2,\,8,7)}{Cl + e^-} \longrightarrow \underset{(2,\,8,\,8)}{Cl^-}$$

(ii) Sulphur atom gains *'two valency electrons'* from its neighbouring atom to complete its last shell. Thus, the negative electrovalency of sulphur is – 2 (S^{--}).

$$\underset{(2,\,8,\,6)}{S + 2e^-} \longrightarrow \underset{(2,\,8,\,8)}{S^{--}}$$

(iii) Phosphorus atom gains *'three valency electrons'* to complete its last shell. Thus, the negative electrovalency of phosphorus is – 3 (P^{---}).

$$\underset{(2,\,8,\,5)}{P + 3e^-} \longrightarrow \underset{(2,\,8,\,8)}{P^{---}}$$

1.25 FORMATION OF ELECTROVALENT COMPOUNDS

1. Formation of Sodium Chloride : Sodium chloride is formed by electrovalent linkage. Sodium atom has an electronic configuration of (2, 8, 1), while chlorine atom has the configuration (2, 8, 7).

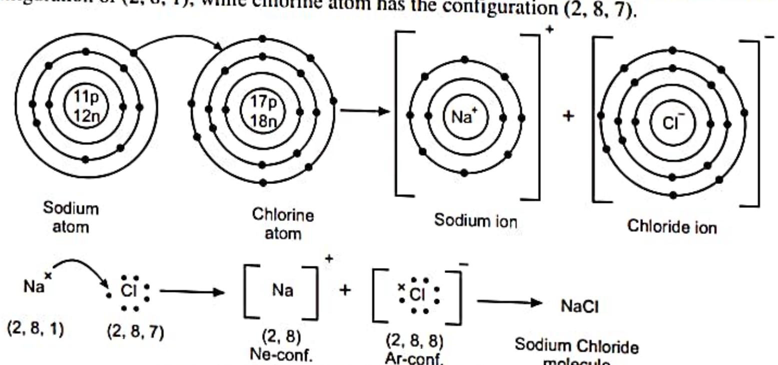
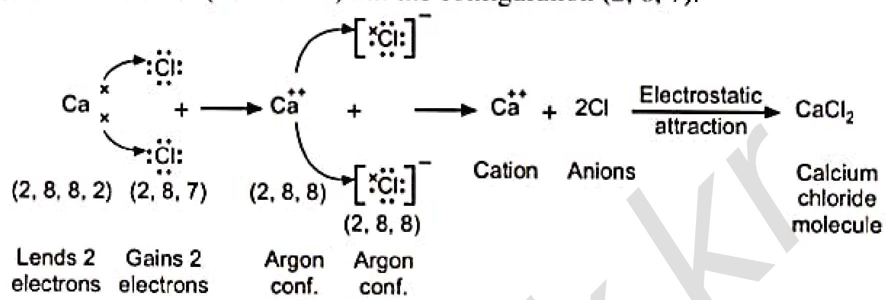


Fig. 1.22 : Formation of Sodium Chloride molecule

During their combination, Na–atom losses its one valency electron and acquires a unit +ve charge and attains a stable configuration (2, 8) as that of neon. The electron lossed by Na–atom is gained by Cl atom, which acquires a unit –ve charge and attains a stable configuration (2, 8, 8) as that of argon. These charged ions (Na^+ and Cl^- ions) thus produced are bound together with electrostatic forces of attraction and produce apparently neutral molecule of sodium chloride (NaCl).

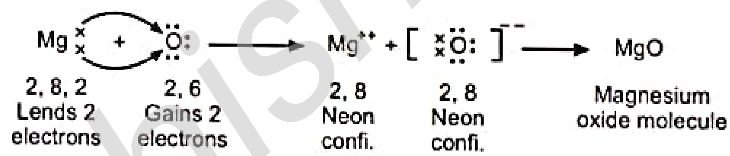
Such a union of atoms which takes place by the loss and gain of electrons is called *'electrovalent linkage'*.

2. Formation of Calcium Chloride Molecule ($CaCl_2$) : A molecule of calcium chloride ($CaCl_2$) consists of one atom of calcium and two atoms of chlorine. Calcium atom (At. No. 20) has the electronic configuration (2, 8, 8, 2), while each chlorine atom (At. No. 17) has the configuration (2, 8, 7).



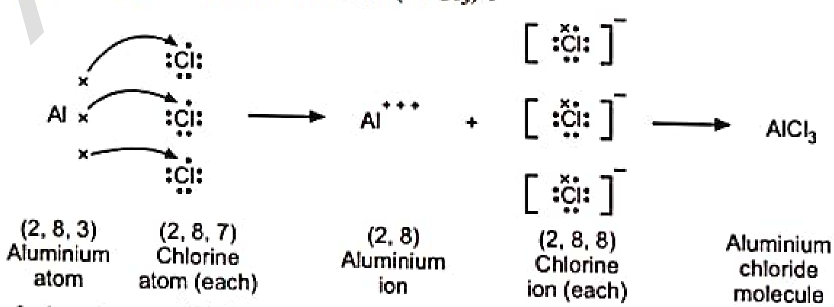
In the formation of calcium chloride molecule, 2 electrons are transferred from calcium atom to two chlorine atoms. By the loss of two electrons, the Ca atom acquires +2 charges and converts into Ca^{++} ion and attains a stable configuration of nearest inert gas element argon. Similarly, two Cl atoms gain one electron each and acquire –1 charge and form 2 Cl^- ions. These oppositely charged ions (Ca^{++} and 2 Cl^-) combine together by the forces of electrostatic attraction to form apparently neutral molecule of $CaCl_2$.

3. Formation of Magnesium Oxide : The formation of magnesium oxide can be diagrammatically represented as follows :



In the formation of magnesium oxide, 2 electrons are transferred from magnesium atom to oxygen atom. By the loss of 2 electrons, the magnesium atom acquires +2 charge and attains the stable configuration of neon gas. On the other hand, oxygen atom acquires –2 charge by the gain of 2 electrons from magnesium atom and attains the stable configuration of neon gas. These two equal and oppositely charged ions (Mg^{++} and O^{--}) unite together by the forces of electrostatic attraction to form apparently neutral molecule of magnesium oxide. Thus, MgO is formed by electrovalency.

4. Formation of Aluminium Chloride Molecule ($AlCl_3$) :



In the formation of aluminium chloride molecule, 3 electrons are transferred from aluminium atom to three chlorine atoms. By the loss of 3 electrons, the aluminium atom acquires +3 charges and attains the stable configuration of the nearest inert gas element neon. On the other hand, each chlorine atom gains one electron from Al atom and acquires –1 charge and attains the stable configuration that of argon. These two equal and oppositely charged ions (Al^{+++} and 3 Cl^-) combine together by the force of electrostatic attraction to form apparently neutral molecule of aluminium chloride ($AlCl_3$).

Properties of Electrovalent Compounds :

1. Electrovalent or ionic compounds are soluble in polar solvents like water, and are insoluble in non-polar solvents like benzene, carbon disulphide and carbon tetra chloride.
2. Electrovalent compounds are ionised in the fused as well as in solution state, hence they conduct electricity.
3. Electrovalent compounds have high melting and boiling points.
4. These are mostly inorganic compounds.

1.26 COVALENCY

"The valency obtained by the mutual sharing of electrons between the similar or dissimilar atoms so as to complete their last orbits is called co-valency or covalent bond." The compounds formed by sharing of electrons between the atoms are called covalent compounds.

It should be noted that in covalent bond (or linkage), no electrical charges are developed as there is no loss and gain of electrons. Here only sharing of electrons takes place. Sharing of electrons occurs in the following ways :

(a) When a covalent compound is formed by mutual sharing of *'one pair'* of electrons between the two atoms, it is called *'single covalent bond'* (–).

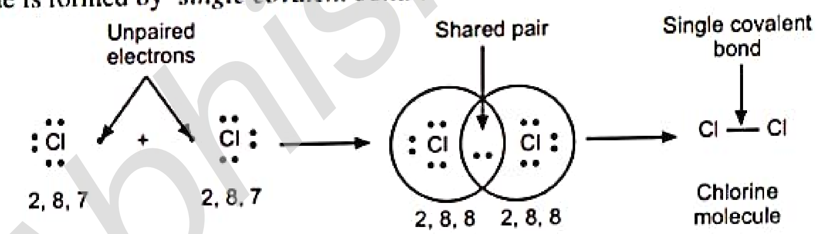
(b) When the covalent compound is formed by mutual sharing of *'two pairs'* of electrons between the two atoms, it is called *'double covalent bond'* (=).

(c) When the covalent compound is formed by mutual sharing of *'three pairs'* of electrons between the two atoms, it is called *'triple covalent bond'* (≡).

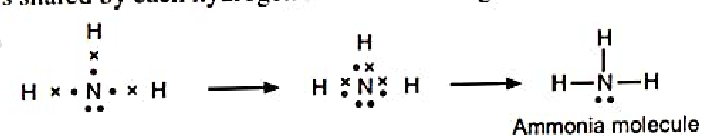
1.27 FORMATION OF COVALENT COMPOUNDS

1. **Single Covalent bond (–) :**

(a) Formation of chlorine molecule (Cl_2) : In the formation of chlorine molecule, each atom of chlorine contains 7 valency electrons. So each chlorine atom is in short of one electron to complete the octet. Thus, a molecule of chlorine is formed by *'single covalent bond'*.

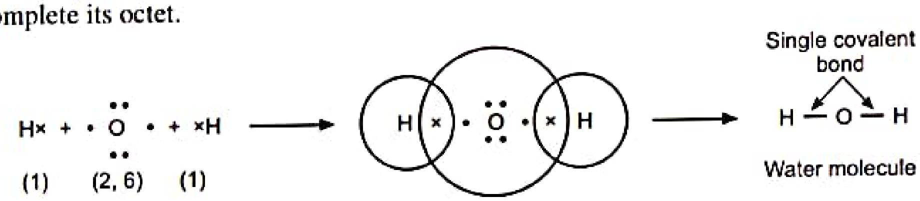


(b) Formation of ammonia molecule (NH_3) : In the formation of ammonia molecule, three atoms of hydrogen (1) combine with one atom of nitrogen (2, 5). Each hydrogen atom contributes one electron with a atom of nitrogen so that a pair of electrons is shared by each hydrogen atom with nitrogen atom.



Thus, nitrogen atom acquires an octet and hydrogen atoms acquire duplets, hence the molecule of ammonia becomes stable.

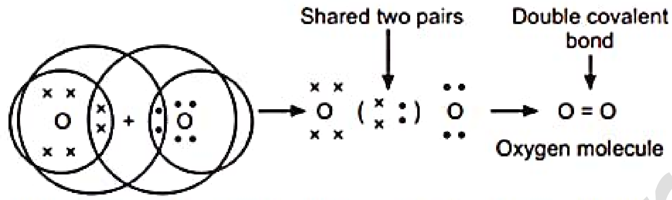
(c) Formation of water molecule (H_2O) : Water molecule (H_2O) contains two atoms of hydrogen and one atom of oxygen. Each hydrogen atom is in short of 1 electron to complete its duplet and oxygen atoms is in short of 2 electrons to complete its octet.



In the formation of water molecule, oxygen atom completes its octet by sharing two electrons with two hydrogen atoms. Similarly, hydrogen atoms complete their duplet by sharing one electron each with oxygen atom. Thus, two separate single covalent bonds are formed between hydrogen and oxygen atoms.

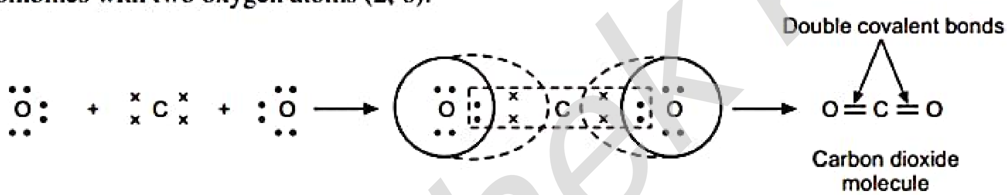
2. Double Covalent bond (=) :

(a) Formation of oxygen molecule (O_2) : Oxygen molecule is diatomic. In the formation of oxygen molecule, each atom of oxygen (2, 6) contains 6 valency electrons. Each oxygen atom is in short of 2 electrons to complete its octet.



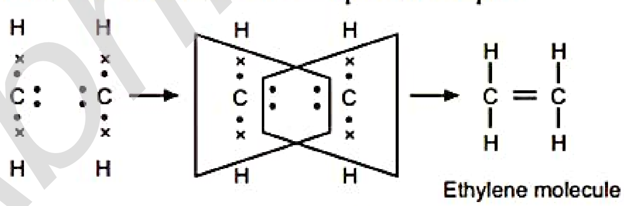
Thus, oxygen molecule is formed by sharing two pairs of electrons between two atoms of oxygen by completing the octet of each. Two shared pairs form a double covalent bond.

(b) Formation of carbon dioxide molecule (CO_2) : In the formation of carbon dioxide molecule, one carbon atom (2, 4) combines with two oxygen atoms (2, 6).



Carbon atom is in short of 4 electrons and each oxygen atom is in short of 2 electrons to complete their octets. Hence, carbon dioxide molecule is formed by two oxygen atoms sharing two pairs of electrons each with a carbon atom. Thus, two double covalent bonds are formed in the formation of CO_2 molecule.

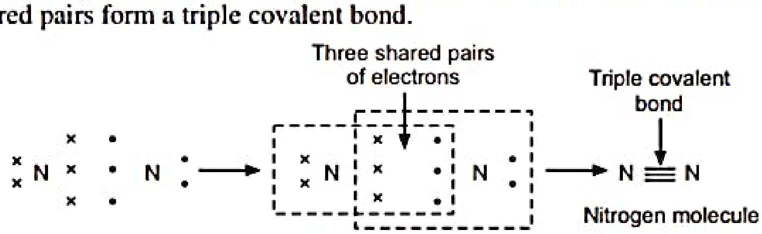
(c) Formation of ethylene molecule (C_2H_4) : The ethylene molecule is formed by the combination of two atoms of carbon (2, 4) and four atoms of hydrogen (1). Each carbon atom is in short of 4 electrons to complete the octet and each hydrogen atom is in short of 1 electron to complete the duplet.



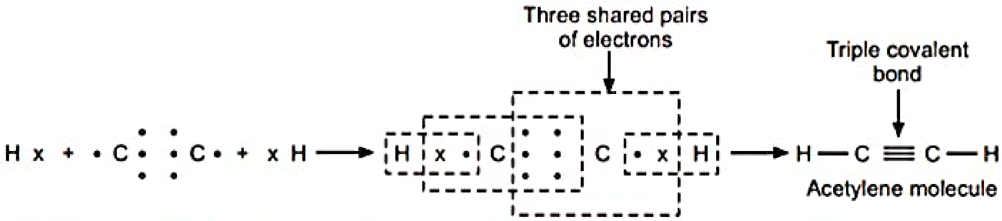
The ethylene molecule is formed by sharing two pairs of electrons between two carbon atoms and by sharing one electron with each of four hydrogen atoms. Thus, the two shared pairs form a double covalent bond between two carbon atoms in ethylene molecule.

3. Triple Covalent bond (≡) :

(a) Formation of nitrogen molecule (N_2) : Nitrogen molecule is diatomic. Each nitrogen atom (2, 5) is in short of 3 electrons to complete the octet. So each nitrogen atom contributes 3 electrons for sharing. Thus, nitrogen molecule is formed by sharing three pairs of electrons between two atoms of nitrogen and hence completing the octet of each. Three shared pairs form a triple covalent bond.



(b) Formation of acetylene molecule (C_2H_2) : Acetylene molecule consists of two atoms of carbon (2, 4) and two atoms of hydrogen (1). Each atom of carbon is in short of 4 electrons to complete the octet, while each hydrogen atom is in short of 1 electron to complete the duplet.



Thus, acetylene molecule is formed by sharing three pairs of electrons i.e. forming triple covalent bond between two carbon atoms and one pair of electrons i.e. single covalent bond with each hydrogen atom.

Properties of covalent compounds :

1. Covalent compounds are generally insoluble in water (polar solvent), but readily soluble in organic solvents such as benzene, carbon tetrachloride (non-polar solvents).
2. Covalent compounds do not ionise when dissolved in water. Thus, they are non-polar in nature. Hence, they do not conduct electricity.
3. Covalent compounds are generally volatile in nature and usually have low melting and boiling points.
4. These are generally organic compounds.

1.28 DISTINCTION BETWEEN ELECTROVALENT AND COVALENT COMPOUNDS

Sr. No.	Electrovalent Compounds	Sr. No.	Covalent Compounds
1.	These are formed by loss and gain of electrons between dissimilar atoms.	1.	These are formed by mutual sharing of electrons between similar or dissimilar atoms.
2.	They are found to exist in the form of ions even in the solid state.	2.	They are not found to exist in the form of ions in the solid or liquid state.
3.	These are polar compounds. Thus, when melted or dissolved in a solvent, they are ionised and hence conduct electric current.	3.	These are non-polar compounds. Thus, when melted or dissolved in a solvent, they are generally not ionised and hence do not conduct electric current.
4.	They possess comparatively high melting and boiling points.	4.	They possess comparatively low melting and boiling points.
5.	They are generally non-volatile and insoluble in organic solvents.	5.	They are generally volatile and usually soluble in organic solvents.

EXERCISE

1. Write short answers of the following : [2 marks]
   (a) Why is an atom electrically neutral ?
   (b) (i) The nucleus consists of 9 protons and 10 neutrons. What should be its atomic weight and electrochemical nature ?
   (ii) For an element having atomic number 9 and mass number 19, write number of protons and neutrons in it.
   (c) State the number of sub-shells in K, L, M, N shells.
   (d) State the relation between atomic number, mass number and neutrons in an atom.
   (e) Define atomic orbital.
   (f) Why is the nucleus of an atom positively charged ?
   (g) Calculate the atomic number and atomic mass number of an atom containing 19 electrons and 20 neutrons.